Document Control

- Document Title: Business Requirements Document (BRD)
- System Name: Application

Executive Summary

- The BRD outlines business requirements for ingesting customer and order CSV data into Delta tables, performing cleansing, computing TotalAmount, SCD Type 2 joins, producing summary and aggregate tables, and operationalizing the pipeline using Delta Live Tables for the Application system.

Business Objectives

- Load source CSVs for customer and order data into Delta tables
- Cleanse data by removing null values and duplicates
- Enrich orders with computed TotalAmount (PricePerUnit × Qty)
- Join customer and order for an ordersummary table with SCD Type 2 history
- Create daily customer spend aggregates in customeraggregatespend
- Leverage Delta Live Tables for implementation

In-Scope

- Ingestion from defined file system locations
- Delta table creation: customer, order, ordersummary, customeraggregatespend
- Data cleansing per requirements
- TotalAmount computation in orders
- SCD Type 2 join logic in ordersummary
- Aggregation for customeraggregatespend
- Implementation via Delta Live Tables

Out-of-Scope

- Data sources or fields not specified in requirements
- Security, access controls, SLAs, monitoring, alerting
- Performance tuning, scheduling, orchestration, DevOps details
- Data quality metrics beyond listed cleansing
- User interfaces, reporting, analytics layers

Stakeholders

- Information not present in requirements input

Business Requirements

BR-APP-001 — Source CSV Ingestion

- Description: Load customer and order CSV files into respective Delta tables from specified paths
- Business Value: Enables downstream data processing and analytics
- Priority: High

BR-APP-002 — Schema Enforcement

- Description: Enforce defined schema for customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId) tables
- Business Value: Ensures data consistency and quality
- Priority: High

BR-APP-003 — Data Cleansing

- Description: Remove "Null"/Null literals and duplicate records from customer and order tables
- Business Value: Improves data accuracy and trustworthiness
- Priority: High

BR-APP-004 — Computed Column TotalAmount

- Description: Add computed column TotalAmount in order table as (PricePerUnit × Qty)
- Business Value: Supports financial calculations and reporting
- Priority: Medium

BR-APP-005 — Ordersummary Table Creation

- Description: Create ordersummary table by joining customer and order on CustId, maintaining SCD Type 2 change history for customer
- Business Value: Provides historical customer-order relationship tracking
- Priority: High

BR-APP-006 — SCD Type 2 Handling

- Description: Apply SCD Type 2 logic to customer dimension—set previous records inactive, update StartDate/EndDate, maintain history
- Business Value: Preserves change audit trail for compliance and analytics
- Priority: High

BR-APP-007 — Aggregate Spend Processing

- Description: Create customeraggregatespend table with daily TotalAmount sums grouped by Name and Date
- Business Value: Enables spend analysis and customer behavior insights

- Priority: Medium

BR-APP-008 — Delta Live Tables Implementation

- Description: Execute all pipeline requirements using Delta Live Tables
- Business Value: Standardizes pipeline management and scalability
- Priority: Medium

Assumptions & Constraints

- Only information from requirements input used
- Delta Live Tables is the mandated implementation tool
- Fields like StartDate, EndDate, Active referenced but not defined explicitly

Success Metrics

- All tables created with defined schemas
- Cleansed, enriched, and history-preserved data
- Accurate daily spend aggregations
- End-to-end flow operational using Delta Live Tables

Risks & Mitigations

- Ambiguity around certain SCD Type 2 attribute definitions
- Mitigation: Raise clarification with stakeholders
- Data quality coverage limited to cleansing steps
- Mitigation: Extend rules if required per future input
- Monitoring, error handling, and performance out of current scope
- Mitigation: Document gaps and address post-MVP

Approval

- Name:
- Title:
- Signature:
- Date:

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into Delta table customer

Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into Delta table order

Appendix B: Target Definitions

	Layer	Table	Columns	Notes
	Raw	customer	CustId, Name, EmailId, Region	Source i
	Raw	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Source i
	Curated	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	SCD Ty
atespend	Gold	customeraggregatespend	Name, TotalAmount, Date	Daily sp

Appendix C: Transformations

sformation	Input	Output	Logic
customer CSV to Delta table	Customer source path	customer	Read and write as Delta
order CSV to Delta table	Order source path	order	Read and write as Delta
ove Nulls and duplicates from customer and order	customer, order	cleansed tables	Filter and deduplicate
pute TotalAmount	order	order	TotalAmount = PricePerUnit × Q
te ordersummary table (SCD Type 2 join)	customer, order	ordersummary	Join on CustId, SCD2 logic
egate spend	ordersummary	customeraggregatespend	sum of TotalAmount grouped by

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-1	Computation	TotalAmount = PricePerUnit × Qty	Requirements input
BR-2	Cleansing	Remove "Null"/Null literals and duplicates from input tables	Requirements input
BR-3	Join	Join customer and order on CustId	Requirements input
BR-4	SCD Type 2	Inactivate previous, activate new, update Start/End dates	Requirements input
BR-5	Aggregation	Group by Name, Date, aggregate sum(TotalAmount)	Requirements input

Appendix E: Process Flow

Step	Description	Dependency
1	Ingest customer and order CSV to Delta tables	None
2	Clean customer and order tables	Step 1
3	Compute TotalAmount in order	Step 1, 2
4	Join customer and order, apply SCD Type 2, create ordersummary	Step 2, 3
5	Aggregate spend, output to customeraggregatespend	Step 4

Appendix F: Entity Relationship Diagram (ERD) Description

- The following ERD captures the key tables and their relationships for the Application:

[customer]

*CustId

Name

EmailId

Region

[order]

*OrderId

ItemName

PricePerUnit

Qty

Date

+CustId

TotalAmount

[ordersummary]

*CustId

Name

EmailId

Region

OrderId

ItemName

PricePerUnit

Qty

Date

[customeraggregatespend]

*Name

TotalAmount

Date

customer *--1 order

customer *--1 ordersummary

ordersummary *--1 customeraggregatespend